

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

SMART METRO TICKET GENERATING SYSTEM

Solution Requirements

1. Purpose and Scope

This document defines the functional and non-functional requirements for the Smart Metro Ticket Generating System (SMTGS). It establishes what the system must do, the quality attributes it must satisfy, and the constraints under which it must operate. The scope covers ticket purchase, fare calculation, payment processing, gate validation, and administrative reporting across all metro stations served by the platform.

1.1 Stakeholders

- Commuters — end users purchasing and using tickets daily.
- Metro Operations Team — manages stations, fare policy and day-to-day service.
- Station Staff — handle gate overrides, assist commuters, resolve disputes.
- Finance / Audit Team — reconciles revenue and ensures regulatory compliance.
- Third-party Payment Providers — UPI switches, card networks, wallet issuers.
- IT / Operations — responsible for platform uptime, security and releases.

1.2 Functional Requirements

Functional requirements describe the observable behaviors the system must provide. The table below lists the core set, each tagged with an ID for traceability and a MoSCoW priority (Must / Should / Could).

ID	Requirement	Priority
FR-01	Commuter can purchase single, round-trip and multi-ride tickets via app, web or kiosk	Must
FR-02	System calculates distance-based or zone-based fare in real time	Must
FR-03	Commuter can pay via UPI, debit/credit card, NFC wallet or smart-card balance	Must
FR-04	System issues a QR code / updates NFC smart card within 3 seconds of payment	Must
FR-05	Turnstile validates ticket and opens gate within 1 second of tap	Must
FR-06	Commuter can recharge smart card / wallet remotely	Should
FR-07	Commuter can request refund/cancellation before journey start	Should
FR-08	Admin can configure fare rules, discounts and station data	Must
FR-09	System generates revenue and ridership reports for operators	Should
FR-10	System sends push/SMS notification on ticket issuance and low balance	Could

2. Requirements Breakdown and Non-Functional Requirements

Beyond individual functional items, requirements are organized into two top-level branches — Functional and Non-Functional — each further decomposed into concrete capability groups. The diagram below visualizes this breakdown alongside the stakeholders and acceptance-criteria sources that govern sign-off.

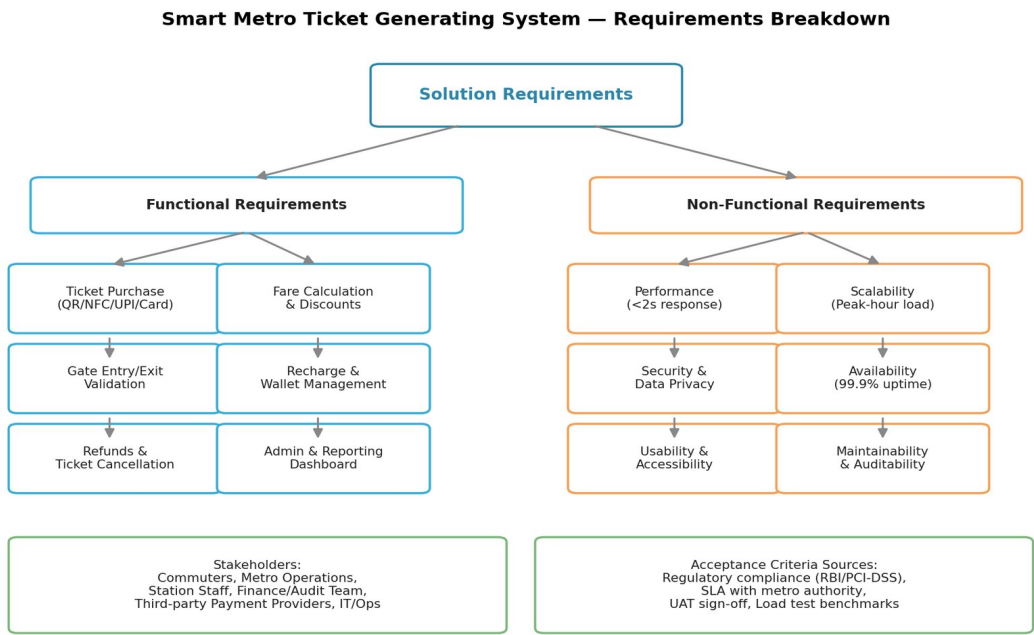


Figure 1: Requirements breakdown structure for the Smart Metro Ticket Generating System

2.1 Non-Functional Requirements

- Performance: 95% of ticket-purchase API calls must complete in under 2 seconds; gate validation under 1 second.
- Scalability: the system must sustain at least 10,000 concurrent transactions during morning/evening peak windows without degradation.
- Security: all payment data encrypted in transit (TLS 1.3) and at rest; PCI-DSS and applicable data-protection compliance.
- Availability: 99.9% uptime for ticketing and gate-validation services, with graceful degradation to offline gate mode during connectivity loss.
- Usability: interfaces must support accessibility guidelines (WCAG 2.1 AA) and be usable in at least two regional languages plus English.
- Maintainability: services must be independently deployable with full audit trails for fare-rule and configuration changes.

3. Requirement Governance, Constraints and Acceptance

Requirements are not static; they are gathered, analyzed, validated and baselined through a controlled workflow so that changes remain traceable throughout the project lifecycle. The workflow below governs how new or changed requirements enter the system of record.

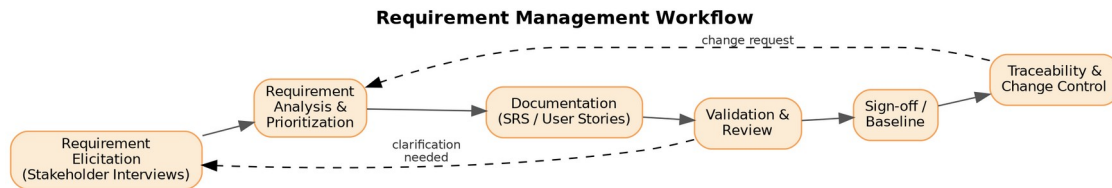


Figure 2: Requirement management workflow, including change-control and re-validation loops

3.1 Assumptions

- Commuters have access to a smartphone or NFC-enabled card for the majority of transactions; cash remains supported only at kiosks.
- Metro operations will supply fare policy and station topology data through the admin configuration interface.
- Third-party payment gateways provide a minimum 99.5% availability SLA.

3.2 Constraints

- The system must integrate with existing turnstile hardware already installed at legacy stations, where full replacement is not budgeted.
- All financial transactions must comply with RBI/PCI-DSS guidelines applicable to the deployment region.
- Initial rollout is constrained to a defined set of pilot stations before network-wide expansion.

3.3 Acceptance Criteria

A requirement is considered accepted only when it has passed User Acceptance Testing (UAT) sign-off from Metro Operations, met the relevant load-test benchmark (where applicable), and been reviewed against regulatory and SLA obligations. Traceability from each functional requirement ID through design, test cases and release notes is maintained for the lifetime of the project.